

# The Evolution of Green Building

*LEED for Homes is just around the corner. Will it fulfill its mission to integrate with existing green building programs while helping out in locations where no such program exists?*

by Jim Gunshinan

**H**ow is the effort to create a national green building program progressing? According to Dennis Creech, executive director of the Southface Energy Institute, in Atlanta, Georgia, pretty well. Creech is a member of a committee that was formed by the United States Green Building Council (USGBC) to develop a national green building program for homes. (See “Who Says a House Is Green?” *HE* Sept/Oct ’03, p.8.) “Efforts to create a national green building standard are very important to transforming the home building market,” Creech says. That’s why he’s investing the time and effort to help create the Leadership in Energy and Environmental Design (LEED) for Homes program. LEED for Homes, or LEED-H, is an effort that USGBC hopes will build on its successful and well-known LEED for commercial buildings program (see “LEED-H Nears Completion”).

However, “home building is local,” says Creech. The LEED for Homes program will use a completely different model than LEED for commercial buildings. “Commercial buildings are built by architects. Homes generally are not,” says Creech. And there are national organizations of architects, such as the American Institute of Architects (AIA), that can push green building values and concepts to their members nationally. There isn’t a similar group for homebuilders. “We have to think of green building in terms of local builders,” says Creech.

Creech has the wisdom of experience. Southface’s EarthCraft House green building program is set to help build



BUILT GREEN WASHINGTON/QUADRANT HOMES

2,000 energy-efficient and sustainable homes in the next year that are easy on the environment and easy on the health of the occupants. The EarthCraft House program has captured 5% of the residential market for new homes in the Atlanta area and wants to increase that to 10% in the next few years. (Atlanta is a large market; it builds about 50,000 new homes a year.) “We’re not interested in boutique homes,” Creech says. “We’re interested in transforming the residential market.”

How can the success of local programs like EarthCraft House translate into creating a national green building program? “There are five key components to most local green building programs that should also be the center of a national green building programs,” says Creech. “These are site selection, including how a site handles water runoff and supports smart

transportation; energy use; water conservation; sustainable materials; and the health of the occupants.” Integrating these basic elements into a national program requires some artistry. “Protecting salmon waterways may be a priority in the Northwest, and that will inform how they handle site selection and water runoff. In Atlanta, we’re concerned with soil erosion from water runoff, and we’re also very concerned with air quality. We have a lot of air conditioners in the area.” Those air conditioners require a lot of electricity, which burdens the electric grid during peak hours in the summer. And the power plants that make that electricity pump a lot of pollution into the air.

Creech also believes that any national program should be based firmly in building science. In other words, green building programs should look at the whole

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## LEED-H Nears Completion

The LEED-H committee was formed in 2000. The committee published draft criteria in 2001, and through 2003 it began the process of building relationships with organizations and agencies that are involved in green building. LEED-H finalized its business model in early 2004. When a LEED-H pilot is launched in April 2005, the key attributes of the program will be sustainable sites, energy and atmosphere, indoor environmental quality, materials and resources, water, and innovation in design and overarching issues (see Table A). There are subheadings under each attribute, with some overlap; home size is important in the design, in materials and resources used, and in the energy use of a home. The pilot program will conclude in April 2006, when the rating system will be launched. (See Table B for the complete timetable.)

According to Ann Edminster, the cochair of LEED-H, while LEED for commercial buildings is aimed at architects, developers, and building owners, LEED-H is aimed at builders. "LEED-H is targeted at builders. If they don't use it, we haven't created a workable tool," she says. Edminster recognizes some logistical and technical challenges yet to be met for LEED-H to be a success. The logistical challenges include program costs, compatibility with other green building programs, how and through whom to deliver the program, and whether to go with a multiple tier rating or a single rating. Addressing the compatibility issue, Edminster says, "USGBC will not actively promote LEED-H in jurisdictions where local programs exist." LEED-H will be promoted in areas without existing green building programs. As for program delivery, LEED-H has developed a business model that is based on the one used by the Residential Energy Services Network (RESNET).

The technical challenges for LEED-H include how to deal with the size of a home. For example, says Edminster, "if a home is 10% over a certain size limit, will the builder have to earn 10% more points in other areas in order to be rated? We don't know yet." Other technical

**Table A. LEED-H Key Attributes**

| Sustainable Sites                               | Materials & Resources                                                                    |
|-------------------------------------------------|------------------------------------------------------------------------------------------|
| Infill & compact development                    | Optimized framing                                                                        |
| Proximity to transit                            | Design for deconstruction & adaptability                                                 |
| Proximity to services                           | Building reuse, deconstruction, & recycling                                              |
| Outdoor living spaces                           | Environmentally preferable materials                                                     |
| Minimizing auto dominance                       | Home size                                                                                |
|                                                 | Envelope durability                                                                      |
|                                                 | Systems integration                                                                      |
| Energy & Atmosphere                             | Water                                                                                    |
| Quality control                                 | Water-efficient fixtures & landscaping                                                   |
| Envelope & mechanical systems performance       | Innovative wastewater technologies                                                       |
| Energy-efficient lighting & appliances          | Reduced energy demand associated with hot water                                          |
| Passive, renewable, & alternative energy        | Drain water heat recovery                                                                |
| Home size                                       | Water catchment                                                                          |
| Systems integration                             |                                                                                          |
| Indoor Environmental Quality                    | Innovation in Design/Overarching Issues                                                  |
| Building envelope moisture control              | Universal design elements                                                                |
| Combustion safety                               | Homeowner/occupant education                                                             |
| Ventilation/air conveyance system effectiveness | Systems integration: integrated solutions to durability, moisture control, IAQ, & energy |
| Vehicle exhaust protection (garage design)      | Home size                                                                                |
| Low-emitting materials                          |                                                                                          |
| Home-landscape interface design                 |                                                                                          |
| Quality control                                 |                                                                                          |
| Systems integration                             |                                                                                          |

**Table B. LEED-H Timetable**

|                                                            |                  |
|------------------------------------------------------------|------------------|
| Hold LHC technical charrette                               | Fall 2004        |
| Finalize draft rating system                               | January 2005     |
| Pre-pilot ("alpha test") rating system with builders       | 1st quarter 2005 |
| Draft pilot reference materials                            | March 2005       |
| Revise rating system as needed based on pre-pilot outcomes | March 2005       |
| Finalize pilot reference materials                         | April 2005       |
| Launch pilot                                               | April 2005       |
| Conclude pilot, launch rating system                       | April 2006       |

challenges include establishing baseline criteria, regional adaptability, and choosing between a prescriptive and a performance-based approach.

How will LEED-H relate to home buyers? "The early adopters are a key component of public education," says Edminster, "along with committed builders and lots of publicity."

While it gets LEED-H off the ground, the USGBC is collaborating with the Smart

Growth and New Urbanist communities to create LEED for Neighborhood Developments, or LEED-ND. LEED-ND will focus its efforts on collaborating with architects and urban planners. The program envisions multiple structures and multiple uses, but of limited scope. Land resources, energy and water use, transportation and mobility, and community and design will be key attributes of the LEED-ND sustainable neighborhoods.

## Three Green Building Programs

Local green building programs are sprouting up across the nation. The three programs described below have each taken a different approach to defining and fostering green building in their communities.

### EarthCraft Expanding

The EarthCraft House green building program out of Atlanta, Georgia, has been in operation since 1999—and it's



EarthCraft's Vickery pilot project is a green-designed community.

been growing steadily since then. EarthCraft has recently expanded into neighboring communities: Savannah, Macon, and Columbus, Georgia, will all have EarthCraft homes in their immediate future. There are even EarthCraft houses planned for other states. A pilot program has been launched in Greenville and Charleston, South Carolina, and contracts are currently under negotiation in Tennessee and Alabama.

The program places an emphasis on providing training and technical assistance to builders. A home must meet energy efficiency requirements and have other green features in order to be labeled an Earthcraft house. All homes are inspected and blower door tested, and the checklist favors features that have the most environmental benefits. Homes must score 150 out of a possible

489 points on the scoring worksheet, with bonus points added for proximity to mass transit, PV and/or solar hot water, and other innovations.

EarthCraft builders must first become members of the Home Builders Association. After that they may join the EarthCraft House program and receive training, design and construction advice, marketing materials, and direct referrals. Incentives for homeowners to build green include reduced closing costs and lower interest rate mortgages, as well as other loan assistance programs. The program has constructed more than 1,700 homes as of this writing.

Builders have a great deal of flexibility in home building through the EarthCraft program. First, builders submit a preliminary worksheet outlining EarthCraft measures they anticipate using on a home. After the building process is under way, builders can adjust these measures, as long as they adhere to the minimum required points. In order for the home to qualify as an EarthCraft house, inspectors must verify that all the measures listed on the worksheet were used during the building process. They conduct visual inspections, perform diagnostic testing, and document results. This includes visually inspecting insulation levels in the attic, doing blower door testing, and securing a copy of the site's erosion control site plan from the builder. All first inspections are free to the builder.

"The home builders have been the core of the program since the beginning," says Jim Hackler, EarthCraft House program director. "It's driven through home builder associations in every area that we're expanding to." With this expansive and integrated program, builders, homeowners, and the environment all stand to benefit. For EarthCraft, it seems as if the future is bright—and big.

### Portland's Comprehensive G/Rated Program



G/Rated, a project of Portland, Oregon's Office of Sustainable Development (OSD), began in 2001 in response to Portland residents' desire for assistance with, education about, and support of green building. Funded through residential and commercial solid-waste fees, grants, and contracts, G/Rated supports several smaller programs, including the Build It Green! residential home tour. This self-



The Rivas-Scott home in Portland, Oregon, is G/Rated.

guided tour through 20 Portland homes is followed by an information fair that gives visitors access to vendors and contractors who are knowledgeable about green building.

Most of the G/Rated program is focused on technical assistance and grants for both residential and commercial building projects. Stephanie Swanson, communications manager at G/Rated, says that G/Rated "works with a lot of other city bureaus and programs in Portland." These include Energy Trust of Oregon, Portland General Electric (PGE) Earth Advantage, and the Bureau of Environmental Services, among others.

G/Rated helps homeowners to solve green building problems, such as how to improve air quality, choose sustainable materials, or install photovoltaic panels.

G/Rated also provides access to green building research, resources, and publications; assistance with loans, tax credits, rebates, and other financial incentives; and training for commercial and residential design construction professionals on green building issues. In addition, the Green Investment Fund, which originally provided 30 homeowners with \$1 million in grant money to encourage the latest in green building, will now provide \$500,000 in grant money annually, beginning this fall.

G/Rated offers one of the nation's most comprehensive green building training series for commercial and residential building design and construction professionals. Launched in February 2003, the 16-week Rethink series explores new ideas, technical training, and the latest in green building innovations and practices. Oregon boasts the most LEED-certified buildings in the country. Swanson says that in the next few years G/Rated wants to continue to "increase professional capacity to make sure expertise in green design and construction is available. We need to continue to push the bar on that." As it stands, Swanson says that the boom in green building practices has paid off for builders, who currently have more work than they can handle.

Other program highlights include the Case Studies section of the G/Rated Web site, which highlights successful green building projects in both residential and commercial categories. These summaries serve as guides for builders, business owners, or homeowners interested in adding green components to their projects. The Web site also provides information about other green programs in Portland, including LEED, the PGE Earth Advantage program, and Energy Star, among many others. A comprehensive list of financial incentives for green building, for both commercial and residential buildings, ensures an even wider public interest in green building practices.

## Frisco Makes It Mandatory

In May 2001, Frisco, Texas, became the first city in the United States to adopt a mandatory residential green building program. Jeff Witt, division manager in the Comprehensive Environmental Planning Office, says that Frisco's building community insisted on the green building program's mandatory status. "The builders told us, 'If you want to



Frisco homes are Energy Star current.

make this a level playing field, you have to make it mandatory."

There has been overwhelming support for the program from both home builders and homeowners. In fact, Witt says that people have been known to buy a home specifically in Frisco (located 30 miles north of Dallas/Fort Worth) "because they have assurance that it will be energy efficient." All homes are performance-tested through HERS.

Frisco's green building program mandates Energy Star program requirements as the minimum building standard for new homes. Thus far, 2,340 homes have been built, with 8,000 lots in the queue. Frisco's green building program focuses on waste reduction, pollution reduction, water conservation, energy conservation, and sustainable development. The city's minimum green building standards revolve around energy efficiency, water

conservation, indoor air quality, and waste reduction. Energy efficiency must be at least equal to Energy Star requirements. Water conservation is emphasized through drought-tolerant landscaping, and by providing information about xeriscaping and water conservation. Irrigation systems must be zoned and must include a rain and freeze sensor, and all bedding areas must be mulched. Indoor air quality requirements include such measures as insulating living areas to R-11, installing CO detectors, sealing all joints of air distribution systems with duct mastic, and providing exterior ventilation systems installed per ASHRAE standards.

Waste is reduced by providing recycling center and compost options, by having builders follow construction waste reduction/reuse plans, and by recycling construction waste, such as brick and wood.

Frisco's Web site has a comprehensive outline of its green building program, with resources for both homeowners and home builders. The city also sponsors a green home fair each year, with people coming from throughout Texas and neighboring states to participate and learn about green building practices.

—Elka Karl

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## For more information:

To learn more about the EarthCraft House program, go to [www.earthcrafthouse.com](http://www.earthcrafthouse.com).

To learn more about G/Rated, go to [www.green-rated.org](http://www.green-rated.org).

To learn more about Frisco's green building program, go to [www.ci.frisco.tx.us/developmentsvcs/greenbuilding/greenbuilding\\_home.htm](http://www.ci.frisco.tx.us/developmentsvcs/greenbuilding/greenbuilding_home.htm).

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Built Green Colorado in Denver is one of the oldest and largest green building programs in the United States.

house as a system. Greg Thomas, president of Performance Systems Development, a consulting firm specializing in home performance and energy efficiency, and president of Performance Systems Contracting, a contracting company—both in Ithaca, New York—echoes Creech's concerns. "Green has become a popular label, and it seems more holistic than home performance, but the green label is not controlled," says Thomas. "Some green building programs place an emphasis on using sustainable materials and are not as concerned with energy efficiency." For Creech and Thomas, green building must integrate whole-house home performance.

And any national green building program should make the health of the occupants a top priority. For example, EarthCraft builders earn points for building according to the guidelines of the American Lung Association's Health House program. "There are other healthy house programs out there. A national program should build on these existing programs," says Creech.

Local programs are successful when they work in tandem with other local, state, and national home building stakeholders. (For examples of successful local programs, see "Three Green Building Programs.") The National Association of Home Builders (NAHB)

sponsors a program for tree conservation called Building with Trees. Atlanta builders earn points toward receiving an EarthCraft House designation by following the NAHB tree program. Southface also works in cooperation with the national Energy Star program, the Atlanta Home Builders Association, and other partners.

Perhaps Creech is strongest in insisting that a national green building program be market based. "The best green building programs won't stop pollution unless houses are built and sold," he says, and "with green building programs, the technical issues are the easiest to solve. Achieving market acceptance is the hardest issue." Creech thinks that a national program should set a standard high enough to contribute meaningful benefits for the environment and for the health and comfort of occupants, but low enough so that mainstream builders can understand the green building concepts, implement them, and then sell the house at a profit. Most of the builders in Georgia come from small construction companies that build 10 to 200 homes a year. These builders don't have the luxury of investing in much training and research. Green building measures must fit easily into their work flow.

Thomas has some concern about setting the bar and leaving it there. If the bar is set low enough to achieve widespread initial success, then how can a green building program raise it after that? "Does it stop with education and getting everyone on board?" Thomas asks. Thomas recommends that national green building requirements be raised over time, and that case studies show the national green building community how to raise the bar.

A national green building program has to respect local climates, local topography, and local building codes. Creech uses Energy Star Homes as an example. "The building codes in Georgia are pretty stringent compared to the codes across the border in Tennessee. It's easier for a builder in Georgia to meet the Energy Star requirements." But that means that the people who build across the border may not be as motivated to build to the Energy Star standard—it's too much of a push. Energy Star takes this difference into account by adapting its standards and requirements by region, and by giving builders some flexibility in how they meet the Energy Star energy efficiency standards. Any national green building program must do the same.

A national green building program that focuses on the basics of site selection, water conservation, energy efficiency, sustainable materials use, and indoor environmental quality; that is market based; and that respects local differences in setting its standards will be a success. Success, for Creech, comes down to market transformation. "That's number one for me," he says. A national program will have the reach to bring in new players to green building. A national program can support the growth of smaller and less developed local green building programs. And a national program can increase the visibility of the existing programs. All that adds up to more healthy and energy-efficient homes; less pollution; and cleaner water for salmon and other creatures.

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